Autonomous Power Routing in Stochastic Smart Grids

Project explainer — what was built, how it works, and what the results mean

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This document is the companion to the A3 poster. The poster has room for conclusions only; this explains the reasoning behind them — what the problem is, why each design choice was made, how the experiment was kept honest, and precisely what the numbers do and do not establish. It is written to be readable by someone who has not seen the code.

1. THE PROBLEM

1.1 What a grid operator is actually deciding

A power network used to be simple in one direction: a few large generators pushed power out to many consumers. Rooftop solar, wind and storage changed that. Every substation is now a source as well as a sink, so the **net imbalance** at a node — generation minus load — swings between surplus and deficit several times a day.

Whenever that imbalance is non-zero, an operator has exactly three options:

- **Curtail** generation — throw away clean energy that has already been produced.
- **Shed** load — cut customers off.
- **Route** power to a neighbouring node that is in the opposite condition.

Routing is by a wide margin the cheapest of the three, which makes “how much power do I send, and along which line?” the decision worth automating. Every megawatt-hour that routing fails to resolve has to be absorbed by one of the other two, both of which cost money and, in the case of reserve generation, carbon.

1.2 Why it is a hard decision rather than an obvious one

Three properties make this a sequential decision problem rather than a lookup table:

- **It is coupled.** A transfer that resolves the imbalance at one node changes the loading on a line elsewhere, and can push that line towards its thermal limit. There is no purely local optimum.
- **It is stochastic and autocorrelated.** The imbalance is weather-dependent, has a strong daily period, and is subject to rare disturbances. An action that is optimal for the current hour may leave the grid badly placed for the next.
- **It is constrained.** Lines have finite capacity, transfers incur I^2R losses, and actuators cannot move instantaneously — they have slew limits.

1.3 The research questions

	Question	Status in this project
RQ1	How much imbalance can a centralized Transformer-DQN remove, measured against an attainable upper bound?	**Answered.** 23.62% of 24.38% attainable.

RQ2	How does a graph-based decentralized controller behave on the same grid?	**Partially.** Implementation and environment verified; performance evaluation incomplete.
RQ3	Can a safety cost keep decentralized learning inside physical limits?	**Partially.** Zero safety cost and zero violations over 5 CPO seeds × 150 episodes.
RQ4	Does it transfer to a larger topology unchanged?	**Not addressed.** Future work.

1.4 The gap this project targets

The contribution here is not a new algorithm. It is methodological, and it comes from a recurring weakness in how results in this area are reported:

- Percentage improvements are published without an attainable upper bound, which makes them uninterpretable — 20% might be excellent or might be a third of what was available.
- Trivial baselines are omitted. On a task where “do nothing” is a legal action and dominates the label distribution, raw accuracy flatters a policy that has learned nothing.
- Centralized and decentralized controllers are usually evaluated in separate papers on separate simulators, so the cost of the information gap between the two cannot be isolated.

This project therefore puts both paradigms on **one simulator** with **one evaluation harness**, and reports every controller against trivial baselines and, where one exists, a computable ceiling.

2. THE SIMULATOR

2.1 Why synthetic data

No external dataset is used. That is a deliberate trade, and it cuts both ways. What is given up is realism — nothing here is validated against a physical network. What is bought is a **known ground truth**: the optimal action at every hour can be computed exactly, unlimited unseen hours can be generated, and the entire study re-runs from a single seed.

2.2 The three nodes

The grid is a triangle: **A** is a city node (solar-peaked), **B** is a suburb (wind- and night-peaked), and **C** is industrial (a twelve-hour load cycle). The phases differ by design, because nodes that peak at the same time have nothing to trade. Measured correlation confirms the headroom exists: A and B are partly complementary ($r = -0.29$), while B and C move together ($r = +0.45$).

2.3 What generates the signal

Mechanism	What it does	Why it is there
Diurnal cycles	Sinusoidal load and generation with node-specific phase	Real grid signals have a dominant 24-hour period
Markov weather	Three-state daily chain (normal / heatwave / storm) rescaling load, solar and wind	Produces regime-switching variance rather than stationary noise
Momentum noise	AR(1) random walk, momentum 0.85	Real forecast errors are autocorrelated; white noise would be too easy
Rare events	Solar outage at A, brown-out at B, industrial demand spike at C	The controller must handle the tail, not just the average day

DC power flow	Impedance, I²R loss, thermal limits	Transfers cost something and lines can be overloaded
Slew-limited actuators	Bounded change per hour, 0.5 response factor	Physical equipment cannot step instantly
N–1 contingencies	A line is removed and the topology changes live	Standard grid reliability criterion

2.4 Verifying the signal is what it claims to be

Before any model was fitted, the generated signal was characterised statistically — because "our simulator is realistic" is an assumption until it is measured.

Test	Result	What it establishes
One-sample t-test on mean imbalance	Rejected at all three nodes ($p < 10^{-19}$)	A structural generation deficit, not noise around zero — there is something to fix
Autocorrelation	lag-1 = +0.95, lag-24 = +0.87 at node A	Strong short-lag persistence *and* a daily echo — justifies a 6-hour window plus an explicit hour feature
Levene test across weather regimes	Rejected ($p < 10^{-45}$ at node A)	Variance genuinely differs by regime — the Markov chain is doing real work
Inter-node correlation	A–B r = –0.29; B–C r = +0.45	Routing between A and B has headroom; B and C move together

The autocorrelation result is the one that drives a design decision: a six-hour window captures most of the short-term structure, and the 24-hour component that the window cannot see is supplied separately as an explicit hour-of-day feature.

2.5 Data quality

Every dataset passes through the same three-stage wrapper — generate, clean, audit — so no model can be handed unvalidated data. The audit reports NaN and infinite counts, duplicate timestamps, physically impossible negative generation or load, and *how often each injected rare event actually fired*. That last check is what turns "the simulator produces storms" from an intention into a verified fact. Cleaning repairs gaps by time interpolation rather than row deletion, because dropping an hour would break the fixed six-hour lookback windows every model depends on.

3. THE CONTROLLERS

3.1 Stage one: a supervised forecaster

Before any control, an LSTM is trained on the plain forecasting task: given the last six hours of imbalance at all three nodes, predict the next hour. This serves two purposes — it proves the six-hour window carries real predictive structure, and its feature layer is later reused as a frozen encoder for a transfer-learning variant of the controller.

Model	MAE (MW)	RMSE (MW)	R^2
LSTM forecaster	5.00	**7.54**	**0.911**
Persistence baseline	6.85	9.57	0.856

That is a 21.2% reduction in RMSE over simply repeating the last observed value. One methodological point matters more than the number: the hold-out split is **chronological, not random**. The sliding windows overlap by

five of their six hours, so a shuffled split would place near-identical windows on both sides of the cut and inflate the score. The reported R^2 is measured on windows that follow the training windows in time and share no hour with them.

3.2 The centralized controller: Transformer-DQN

This is the architecture that produces the headline result.

Observation. A 6 × 3 window — six hours of net imbalance at three nodes, normalised by a fixed scale so that the frozen encoder later sees exactly the distribution it was trained on.

Encoder. The window is projected to 32 dimensions, a learned positional embedding is added, and a four-head self-attention block with a feed-forward layer and residual layer normalisation processes it, then pools over time.

Why attention rather than a recurrent net? Because the signal has two timescales. Attention lets the controller weight the hours in its window by relevance rather than by recency, which suits a signal with both a short-lag component (lag-1 = 0.95) and a daily one (lag-24 = 0.87).

Head. A standard Deep Q-Network: replay buffer, an independently constructed target network, ε-greedy exploration decayed once per episode, discount $\gamma = 0.95$.

Action space. Seven discrete actions — Hold, or move a fixed 20 MW along one of the six directed node pairs (A→B, B→C, C→A, B→A, C→B, A→C).

Reward — the choice worth explaining. The reward is *differential*: total absolute imbalance before the action, minus total absolute imbalance after, minus a cost term on the I^2R loss, scaled down. Written out:

$$S(t) = \Sigma_i \left| G_i(t) - L_i(t) \right|$$

$$\text{reward} = \left(S(\text{before}) - S(\text{after}) - c_{\text{loss}} \cdot I^2R \right) / 10$$

This is literally “the megawatts of grid stress this action removed”. The alternative — rewarding low absolute imbalance — was tested and rejected, because the pre-action total is a constant of the state: it adds a large state-dependent offset that dominates the Q-targets and carries no information about which of the seven actions to choose. The differential form ranks the actions exactly as the oracle would, with far less target noise.

3.3 The greedy oracle, and why it is the most important component

At every state, the oracle brute-forces all seven actions and picks the one that minimises total absolute imbalance afterwards. Because the action space is tiny, this is not an approximation — it is the exact optimum for a single 20 MW transfer in that hour.

It does two jobs. It provides a **ceiling**, so a percentage reduction becomes interpretable. And it provides a **label** for every state, so the controller can also be scored with familiar supervised metrics: accuracy, per-class F1, and Cohen's κ.

One honest qualification: the oracle is *myopic*. It is the best single transfer per hour, not the best policy over a horizon. An agent that sacrifices the current hour to position the grid better for the next could in principle exceed it. “96.9% of the ceiling” should be read as 96.9% of the *greedy* ceiling.

3.4 The decentralized controller: graph multi-agent RL

This architecture throws away the central view deliberately, to measure what that information is worth.

- **One agent per substation.** Each observes only its own imbalance, the hour feature, what its neighbours broadcast, the loading of the line it controls, and the live adjacency matrix — so topology changes, including an N−1 outage, are visible as a change in observation rather than a silent break in assumptions.
- **Graph neural network** for neighbour aggregation. This is the component that would let the approach scale to a grid with more than three nodes without redesigning the controller.
- **Continuous actuators:** a set-point in ±50 MW with a 15 MW-per-hour slew limit and a 0.5 actuator

response, so no line can be driven to capacity in a single step.

- **Two training algorithms:** PPO, and CPO — Constrained Policy Optimization, which enforces a hard safety constraint rather than folding safety into the reward as a penalty.

Why bother, when the centralized controller is nearly optimal? Because on three nodes centralization wins trivially. Decentralization is advocated for real grids on two grounds — it scales with the number of nodes, and it has no single point of failure — and the information each agent gives up is rarely quantified. That quantification is the long-term aim; this project builds and verifies the apparatus for it.

4. HOW THE EXPERIMENT WAS KEPT HONEST

This section is disproportionately long relative to its share of the poster, which reflects where the effort actually went. A large part of the work was spent making the experiment *valid* rather than making it larger.

4.1 Evaluation protocol

- Every controller is scored over the same **240 unseen hours** (ten simulated days), generated from a seed never used in training. The unmanaged reference total is **16,303.73 MW** of accumulated imbalance.
- Within a paradigm, the same seed fixes both the window start and the weather sequence, so no controller gets an easier ten days than another.
- Evaluation is always greedy — the DQN acts through a dedicated greedy path that ignores ε entirely, so residual exploration cannot leak into a reported number.

4.2 Baselines with known answers

Two of the reference policies exist specifically so the harness can be checked against a result known before the experiment runs:

- **Always-Hold** takes no action, so its reduction must be exactly 0.00%. It is.
- **No-control** in the decentralized harness must likewise be exactly 0.00%. It is, to the last reported digit.

Together these prove that the “managed imbalance” figure measures the effect of the actions and nothing else — that no part of the reduction is an artefact of how the window is selected or how totals are accumulated.

A third baseline, **random**, exists to answer a different question: is this a task where any action helps? It scores −14.91%, so no. Acting arbitrarily is substantially worse than doing nothing, which is what makes beating zero a real result.

4.3 Oracle–environment consistency

The oracle and the environment are two independent implementations of the same physics — the oracle as an explicit search over the six transfers, the environment through a delta matrix. If they ever drifted apart, every accuracy and κ figure would be scoring the agent against a different grid from the one it trained on. A dedicated check re-scores every labelled state through the environment's own dynamics and confirms the oracle's label is in fact minimal. Measured agreement: **100%**. This is a precondition for every decision-quality number, not a performance claim.

4.4 Defects found and corrected during implementation

Several of these materially changed the numbers a run produced, and are recorded because the results are not interpretable without knowing they were fixed.

Defect	Consequence if left

The target network was built from the same layer objects as the online network	It was an alias, not a copy — the target update was a no-op and Q-learning bootstrapped off constantly moving targets
ε was decayed inside the replay step (~50× per episode)	Exploration collapsed to its floor within about 3 of 60 episodes — the agent barely explored at all
The forecaster's training data was never split	No way to distinguish a learned backbone from a memorised one
Decentralized agents could not observe the shunt level they control	Their actions were deltas on a hidden state, so the process was not Markov
PPO took a single gradient step per rollout	The importance ratio is exactly 1.0, making the clipped surrogate mathematically inert
Truncated episodes bootstrapped the value target from zero	Taught the critic that the world genuinely ends every 100 hours

4.5 Decisions settled by measurement

Several design choices were tested rather than argued, and two changes that seemed correct in principle were measured, found harmful, and reverted:

- **Bidirectional transmission lines** — reverted. Allowing reverse flow moved the decentralized controller sharply *worse*, because all three nodes are usually in deficit simultaneously and the extra freedom mostly permits importing power that no neighbour has.
- **Differential vs. absolute reward** — adopted (see §3.2).
- **Own line loading as an observation channel** — adopted; without it the decentralized process is not Markov.

5. RESULTS

5.1 Physical performance over 240 unseen hours

Policy	Imbalance reduction	Reading
Greedy oracle (upper bound)	**24.38%**	The ceiling for one 20 MW transfer per hour — nothing can beat this
Transformer-DQN	**23.62%**	**96.9% of the attainable control.** The headline result
TransferAgent (frozen LSTM encoder)	22.00%	Roughly 90% of the ceiling — measurably below the attention agent
Always-Hold	0.00%	Exactly zero by construction — the harness sanity check
Random	−14.91%	Acting arbitrarily is much worse than not acting — this is not a task where any transfer helps

How to read the headline. 23.62% in isolation means nothing. Against a ceiling of 24.38% it means the controller has captured almost all of the control that physically exists in this action space; the remaining headroom is 0.76 percentage points.

5.2 Decision quality against the oracle

Metric	Value	What it means

Oracle agreement (accuracy)	59.8%	The agent picks the myopically optimal action in roughly 3 hours out of 5
Cohen's κ	0.488	Chance-corrected agreement. Always-Hold scores exactly 0.000 on the same metric
Oracle–physics match	100%	Correctness check (§4.3), not a performance figure
Hold's share of oracle labels	39.1%	Hold does not dominate enough for a do-nothing policy to look good

The reason κ is reported alongside accuracy is that Hold is the single most common correct action. A policy that always holds would score a respectable-looking accuracy while contributing no information at all — and κ = 0.000 is the correct reading of such a policy. At κ = 0.488 the controller is in the range conventionally described as moderate-to-substantial agreement.

5.3 The trade-off

The result is not free, and the poster says so. The learned controller switches more often and incurs higher I^2R loss than the oracle does to achieve a comparable reduction. The agent prefers to act in hours where holding would have been marginally better — which costs transmission but little imbalance.

This is a finding rather than a defect: **imbalance reduction and routing efficiency are distinct objectives**. The reward optimised the first with only a light penalty on the second. A deployed system would need both weighted explicitly, and the weighting is a policy decision, not a modelling one.

5.4 Decentralized control: what is and is not claimed

Claimed: the decentralized environment, observations, actuators and safety machinery are implemented and verified. Across 5 CPO seeds × 150 episodes the controller recorded **zero safety cost and zero thermal or voltage violations** — the constraint does what it is meant to do. A classical proportional rule on the same 48-hour reference removes 21.24% of stress for 1.32 MW of loss and no violations, which establishes the reference any learned controller must beat.

Not claimed: that the learned decentralized controller performs well. The PPO-versus-CPO comparison is incomplete, so no decentralized performance figure appears on the poster. An empty cell is preferable to a number that would have to be retracted.

5.5 The transfer-learning result, which went the wrong way

The expectation was that reusing the forecaster's pre-trained encoder would help. It did not: the transfer agent reached 22.00% against the attention agent's 23.62%.

The explanation is worth stating because it generalises. The encoder was pre-trained to **forecast next-hour imbalance**, which is a different objective from **ranking seven actions by their immediate effect**. The features that best predict where the imbalance is going are not necessarily the features that best discriminate between actions. With the encoder frozen, the agent can only learn a small head on top of representations shaped by the wrong objective, and the measured cost of that constraint is roughly 1.5 percentage points of reduction. A pre-trained encoder is not automatically a good control encoder.

6. LIMITATIONS

Stated plainly, because they bound what the results can be used for.

- **The grid is synthetic.** The simulator is physically motivated but uses no real weather, demand or price data. The results characterise the controllers on this process, not on a physical network.

- **The transmission model is idealised.** No voltage or frequency dynamics, no reactive power, and transfers take effect within the same hour.
- **The action space bounds the result.** Exactly one 20 MW transfer per hour is permitted centrally, which is what caps the attainable reduction at 24.38% on this window.
- **Single-seed runs.** Each figure comes from one seeded run, not a mean with a confidence interval, so run-to-run variance is not quantified and small differences between controllers cannot be tested.
- **The oracle is myopic.** It is the best single transfer per hour, not the best policy over a horizon.
- **Modest training budgets.** Neither controller can be claimed to be at convergence.
- **RQ4 is unaddressed.** Nothing here tests transfer to a larger topology.

7. WHAT COMES NEXT

1. **Finish the PPO vs. CPO comparison** against the proportional-rule reference, so the decentralized arm has a defensible performance number rather than an implementation claim.
2. **Repeat every experiment over multiple seeds** and report means with confidence intervals, so differences between controllers can be tested rather than merely observed.
3. **Double-Q learning** to reduce the value overestimation inherent in standard DQN, and a tuned exploration schedule, to test whether the remaining headroom to the oracle can be recovered.
4. **Scale past three nodes** — this is where the decentralized paradigm should begin to pay for itself, and where the centralized action space grows quadratically.
5. **Address the decentralized credit assignment directly:** a shared or mixed reward rather than a purely local one, and a wider per-hour bound so the actuator can track the imbalance.
6. **Extend the physics:** line losses and impedance in more detail, storage as a controllable element, and transfers that persist across hours.
7. **Replace the simulator with real data** eventually, retaining the audit stage so data-quality issues surface explicitly rather than silently.

8. WHAT TO TAKE AWAY

1. A Transformer-based DQN removes **23.62%** of total grid imbalance over 240 unseen hours — **96.9%** of what is physically attainable with a single 20 MW transfer per hour.
2. That claim is only meaningful because a computable ceiling, two trivial baselines and a 100%-verified physics check sit behind it. The methodology is the contribution, more than the architecture.
3. The result is not free: imbalance reduction and routing efficiency are separate objectives, and only the first was optimised.
4. Pre-training does not automatically transfer. A frozen forecasting encoder cost about 1.5 percentage points relative to one trained for the control objective.
5. The decentralized graph-MARL arm has a verified foundation and a working safety constraint — zero violations over five CPO seeds — but no performance claim, and that is stated rather than papered over.

GLOSSARY

Term	Meaning in this project

Net imbalance	Generation minus load at a node, in MW. Positive = surplus, negative = deficit. The quantity every controller minimises in absolute total across the grid
DQN	Deep Q-Network. Learns the expected return of each discrete action and acts greedily on it. Uses a replay buffer and a separate target network for stability
Transformer / self-attention	Architecture that weights each timestep in the input window by learned relevance rather than by recency
Greedy oracle	Brute-force optimal single action per hour. Provides both the performance ceiling and per-state labels
Cohen's κ	Agreement corrected for the agreement expected by chance. Zero means “no better than guessing the class distribution”
PPO	Proximal Policy Optimization. Policy-gradient method for continuous actions; clips the update so a single step cannot move the policy too far
CPO	Constrained Policy Optimization. PPO with an explicit safety constraint rather than a reward penalty
MARL	Multi-Agent Reinforcement Learning. Several independent learners sharing one environment
I^2R loss	Resistive power lost in a line, proportional to the square of the current. Why routing is not free
N–1 contingency	Standard reliability criterion: the grid must cope with any single component failing
Slew limit	Maximum rate at which an actuator can change its set-point per hour
Persistence baseline	Forecast that simply repeats the last observed value. The bar any forecaster must clear

PROJECT FILES

File	Contents
`PBL.ipynb`	Complete implementation: simulator, preprocessing, statistical tests, forecaster, both control paradigms, evaluation harness, consolidated results. Runs top-to-bottom from a fresh kernel
`Smart_Grid_Poster_A3_improved.pdf`	The A3 conference poster
`PBL_Deliverables/PBL_Project_Report.docx`	Full report: methodology, validation, per-class metrics, telemetry tables, references
`PBL_Deliverables/PBL_Project_Presentation.pptx`	Slide deck built from the same figures
`pbl_artifacts/`	Trained forecaster (`grid_forecaster.keras`) and all generated figures — EDA, autocorrelation, forecaster evaluation, learning curves, confusion matrices, routing telemetry
`PBL_Deliverables/References/`	The cited literature as PDFs

Note on figure consistency. Some values differ between the poster and the report — the poster reports 23.62% reduction, a 24.38% ceiling and κ = 0.488, where the report states 23.52%, 24.39% and κ = 0.586; the report also records a completed decentralized PPO evaluation that the poster describes as incomplete. This document follows the poster, as the later artefact. Confirm which run is current before presenting or submitting, and align all three documents on one set of numbers.